

Newman-Penrose Formalism Calculations

Stephani et al., Equation (33.44a_2)

Metric

$$ds^2 = -dt \otimes dt + \Phi^2(t) \left(x^4 U(r)^2 + y^4 U(r)^2 - 4y^3 U(r) V_i(r) + 4x^3 U(r) V_r(r) + 2 \left(A(t, r) U(r) + 2V_r(r)^2 \right) x^2 + 2 \left(x^2 U(r)^2 + 2x U(r) V_r(r) + A(t, r) U(r) + 2V_i(r)^2 \right) y^2 \right. \\ \left. + 4xA(t, r) V_r(r) - 4 \left(x^2 U(r) V_i(r) + 2x V_i(r) V_r(r) + A(t, r) V_i(r) \right) y + A(t, r)^2 \right) dr \otimes dr + \Phi^2(t) dx \otimes dx + \Phi^2(t) dy \otimes dy$$

Coordinates

$$(x^\mu) = (t, r, x, y)$$

Null Tetrad

Covariant components

$$l = \sqrt{\frac{1}{2}} dt + \frac{\Phi(t)}{\sqrt{2}} (A + (x^2 + y^2) U + 2V_r x - 2V_i y) dr$$

$$n = \sqrt{\frac{1}{2}} dt - \frac{\Phi(t)}{\sqrt{2}} (A + (x^2 + y^2) U + 2V_r x - 2V_i y) dr$$

$$m = \left(\sqrt{\frac{1}{2}} \Phi(t) \right) dx + \left(i \sqrt{\frac{1}{2}} \Phi(t) \right) dy$$

$$\bar{m} = \left(\sqrt{\frac{1}{2}} \Phi(t) \right) dx + \left(-i \sqrt{\frac{1}{2}} \Phi(t) \right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\Phi(t) \sqrt{2}}{2 |x^2 U(r) + y^2 U(r) - 2yV_i(r) + 2xV_r(r) + A(t, r)|} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{\Phi(t) \sqrt{2}}{2 |x^2 U(r) + y^2 U(r) - 2yV_i(r) + 2xV_r(r) + A(t, r)|} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2 \Phi(t)} \right) dx + \left(\frac{i \sqrt{2}}{2 \Phi(t)} \right) dy$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2 \Phi(t)} \right) dx + \left(-\frac{i \sqrt{2}}{2 \Phi(t)} \right) dy$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2} \Phi(t)}{2 |x^2 U(r) + y^2 U(r) - 2yV_i(r) + 2xV_r(r) + A(t, r)|} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t - \frac{\sqrt{2} \Phi(t)}{2 |x^2 U(r) + y^2 U(r) - 2yV_i(r) + 2xV_r(r) + A(t, r)|} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2 \Phi(t)} \partial_x + \frac{i \sqrt{2}}{2 \Phi(t)} \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 \Phi(t)} \partial_x - \frac{i \sqrt{2}}{2 \Phi(t)} \partial_y$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} \frac{\partial}{\partial t} \Phi(t)}{2 \Phi(t)}$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & \frac{\sqrt{2}xU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} + \frac{i\sqrt{2}yU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \\ & - \frac{i\sqrt{2}V_i(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} + \frac{\sqrt{2}V_r(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \end{aligned}$$

$$\mu = -\frac{\sqrt{2}\frac{\partial}{\partial t}\Phi(t)}{2\Phi(t)}$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & -\frac{\sqrt{2}xU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} + \frac{i\sqrt{2}yU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \\ & - \frac{i\sqrt{2}V_i(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} - \frac{\sqrt{2}V_r(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \end{aligned}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\begin{aligned} \tau = & -\frac{\sqrt{2}xU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} - \frac{i\sqrt{2}yU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \\ & + \frac{i\sqrt{2}V_i(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} - \frac{\sqrt{2}V_r(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \end{aligned}$$

$$\begin{aligned} \pi = & \frac{\sqrt{2}xU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} - \frac{i\sqrt{2}yU(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \\ & + \frac{i\sqrt{2}V_i(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} + \frac{\sqrt{2}V_r(r)}{2(x^2\Phi(t)U(r) + y^2\Phi(t)U(r) - 2y\Phi(t)V_i(r) + 2x\Phi(t)V_r(r) + A(t,r)\Phi(t))} \end{aligned}$$

$$\begin{aligned}
\epsilon = & - \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))} \\
& - \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))} \\
& + \frac{y V_i(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t)} \\
& - \frac{x V_r(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t)} \\
& - \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))} \\
& - \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))}
\end{aligned}$$

$$\begin{aligned}
\gamma = & \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))} \\
& + \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))} \\
& - \frac{y V_i(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t)} \\
& + \frac{x V_r(r) \frac{\partial}{\partial t} \Phi(t)}{\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t)} \\
& + \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))} \\
& + \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 (\sqrt{2} x^2 \Phi(t) U(r) + \sqrt{2} y^2 \Phi(t) U(r) - 2 \sqrt{2} y \Phi(t) V_i(r) + 2 \sqrt{2} x \Phi(t) V_r(r) + \sqrt{2} A(t, r) \Phi(t))}
\end{aligned}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = & \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{x^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{y^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{y V_i(r) \frac{\partial}{\partial t} \Phi(t)^2}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} + \frac{x V_r(r) \frac{\partial}{\partial t} \Phi(t)^2}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} \\
& + \frac{y \Phi(t) V_i(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} - \frac{x \Phi(t) V_r(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} \\
& + \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{U(r)}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2}
\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = & \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{x^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{y^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{y V_i(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{x V_r(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{y \Phi(t) V_i(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{x \Phi(t) V_r(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)}
\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned} \Phi_{22} = & \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & + \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & - \frac{x^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & - \frac{y^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & - \frac{y V_i(r) \frac{\partial}{\partial t} \Phi(t)^2}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} + \frac{x V_r(r) \frac{\partial}{\partial t} \Phi(t)^2}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} \\ & + \frac{y \Phi(t) V_i(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} - \frac{x \Phi(t) V_r(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} \\ & + \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & + \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & - \frac{A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & - \frac{U(r)}{x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2} \end{aligned}$$

$$\begin{aligned}
\Lambda = & \frac{x^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{y^2 U(r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{x^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{y^2 \Phi(t) U(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{y V_i(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{x V_r(r) \frac{\partial}{\partial t} \Phi(t)^2}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{y \Phi(t) V_i(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{x \Phi(t) V_r(r) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{2 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{12 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{3 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{A(t, r) \frac{\partial}{\partial t} \Phi(t)^2}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& + \frac{A(t, r) \Phi(t) \frac{\partial^2}{(\partial t)^2} \Phi(t)}{4 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\
& - \frac{U(r)}{3 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)}
\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & -\frac{\Phi(t)^2 \frac{\partial^2}{(\partial t)^2} A(t, r)}{6 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & -\frac{\Phi(t) \frac{\partial}{\partial t} A(t, r) \frac{\partial}{\partial t} \Phi(t)}{6 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)} \\ & -\frac{U(r)}{3 \left(x^2 \Phi(t)^2 U(r) + y^2 \Phi(t)^2 U(r) - 2 y \Phi(t)^2 V_i(r) + 2 x \Phi(t)^2 V_r(r) + A(t, r) \Phi(t)^2 \right)}\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"